DATA ANALYST CASE STUDY

Understanding the Child in Need (CIN) Population

An exploratory analysis of vulnerability, repeat episodes, and age patterns for a UK Local Authority Children's Services department

Prepared for Social Finance • Monday 27 July 2026

AGENDA

What this presentation covers

01

The population

Who are the children in this dataset, and how many experience a Child in Need episode?

02

One episode vs. three or more

How do children with a single CIN episode differ from those with repeated episodes?

03

The role of age

Do these differences show up the same way across early years, primary, secondary and post-16?

04

Data, limits & next steps

What would sharpen this analysis further — and what should we ask the LA?

CONTEXT

Why this matters to the Local Authority

Children's Services already has an intuitive, frontline sense of how vulnerable children behave. This analysis is about turning that intuition into empirical, fact-based evidence — so the Local Authority can act earlier and more precisely to support this population.

WHAT IS A CIN EPISODE?

A 'Child in Need' (CIN) episode is a period during which Children's Services has formally determined a child is acutely vulnerable and requires action — short of statutory child protection.

THE TWO QUESTIONS WE FOCUS ON

1. What distinguishes a child with exactly 1 CIN episode from one with 3+ episodes?
2. How do those distinctions change with the child's age?

THE DATA

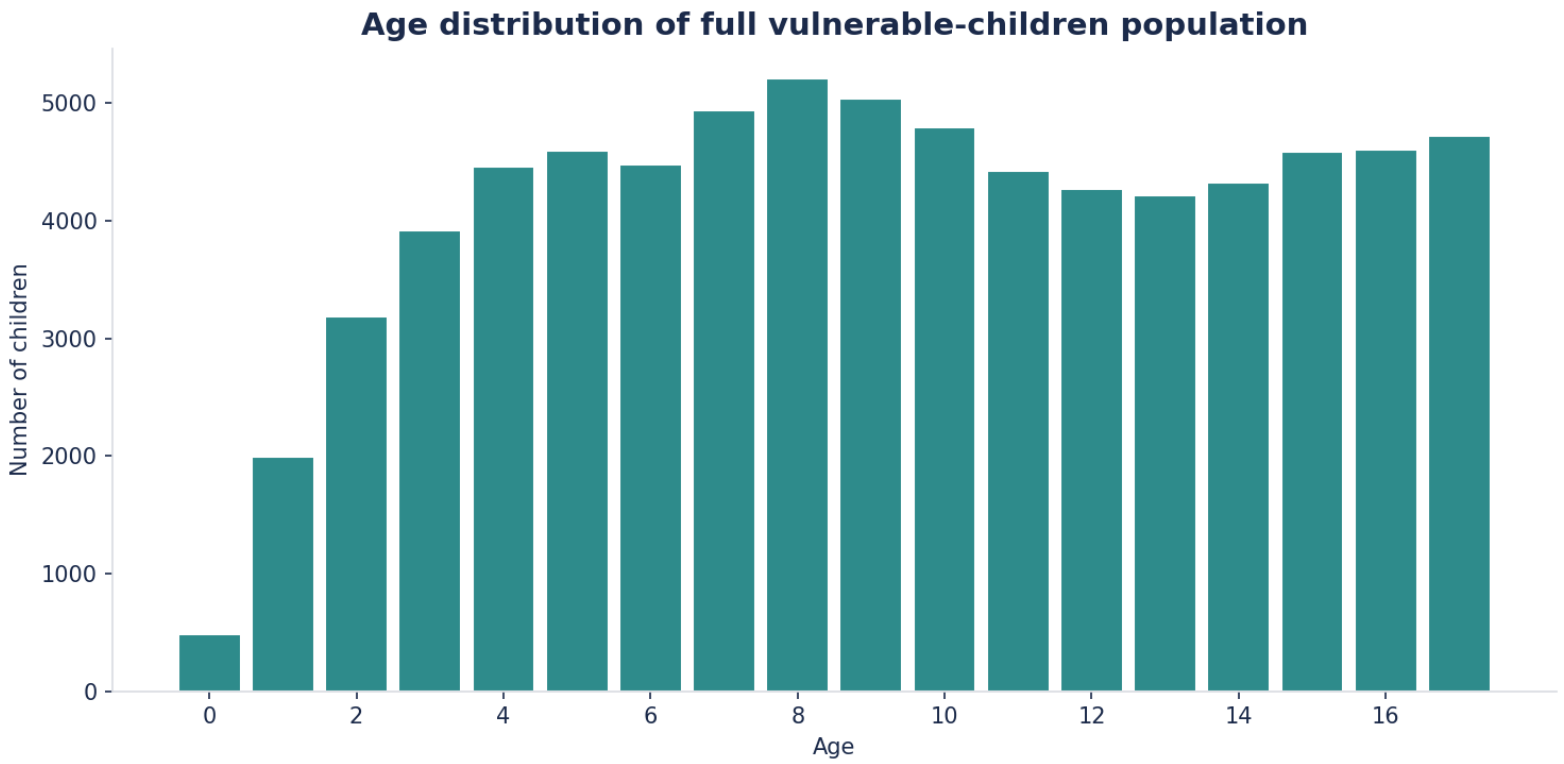
One Local Authority, 74,247 children, 28 fields

74,247	28	30.3%	0
children in the dataset	fields per child	have ≥1 CIN episode	duplicate or missing records

A NOTE ON DATA QUALITY

- DWP, AdultOffender and NEET are 0 for every one of the 74,247 children — these fields appear unpopulated rather than genuinely absent, and are excluded from our conclusions.
- "Exclusion" (ever excluded) doesn't fully nest inside "Fixed-period exclusion" as expected — 2,520 children have a fixed-period exclusion flag but no exclusion flag. Worth clarifying the field definitions with the LA.
- The data is a point-in-time snapshot with no dates — we can describe association, not sequence or causality.

A broad population, spanning every age from birth to 17

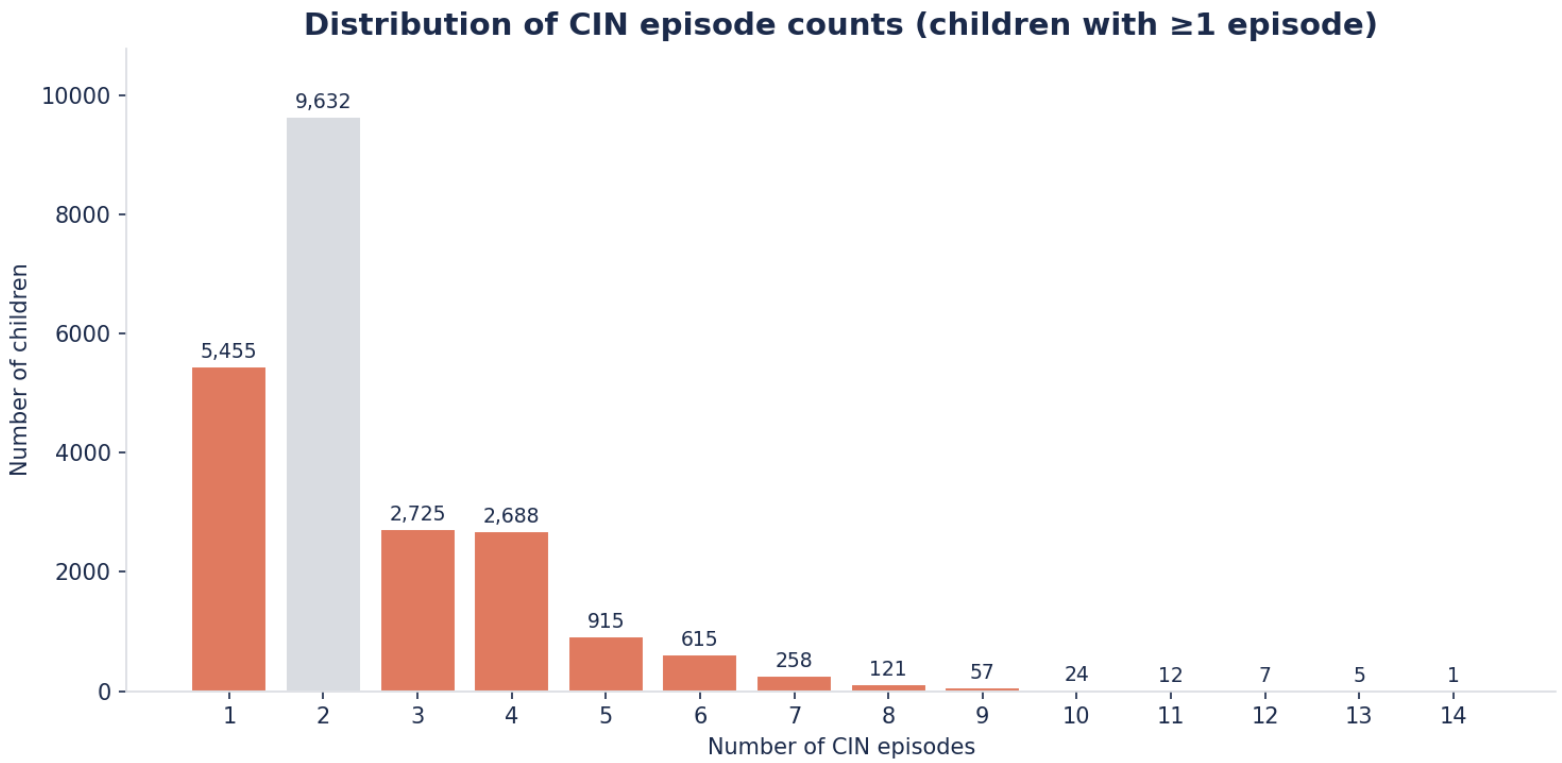


READING THE SHAPE

Numbers rise through early childhood, peak around age 13-14, then fall off sharply after 16 as young people age out of children's services.

This matters for our later age-based comparisons: the 16-17 group is naturally the smallest, so small differences there carry more statistical noise.

Most CIN children have 1–2 episodes — a minority recur often



KEY NUMBERS

5,455

children with exactly 1 episode

7,428

children with 3+ episodes — our two comparison groups for the rest of this deck

Two groups, one underlying question: single crisis, or a pattern?

GROUP A

Exactly 1 episode

n = 5,455

A child who needed formal support once, and did not re-enter the CIN process again — potentially a single, resolved crisis.

GROUP B

3 or more episodes

n = 7,428

A child who has repeatedly re-entered the CIN process — suggesting recurring or entrenched need rather than a one-off event.

Repeat-episode children show more associated indicators across the board

BIGGEST GAPS

+15.5pp

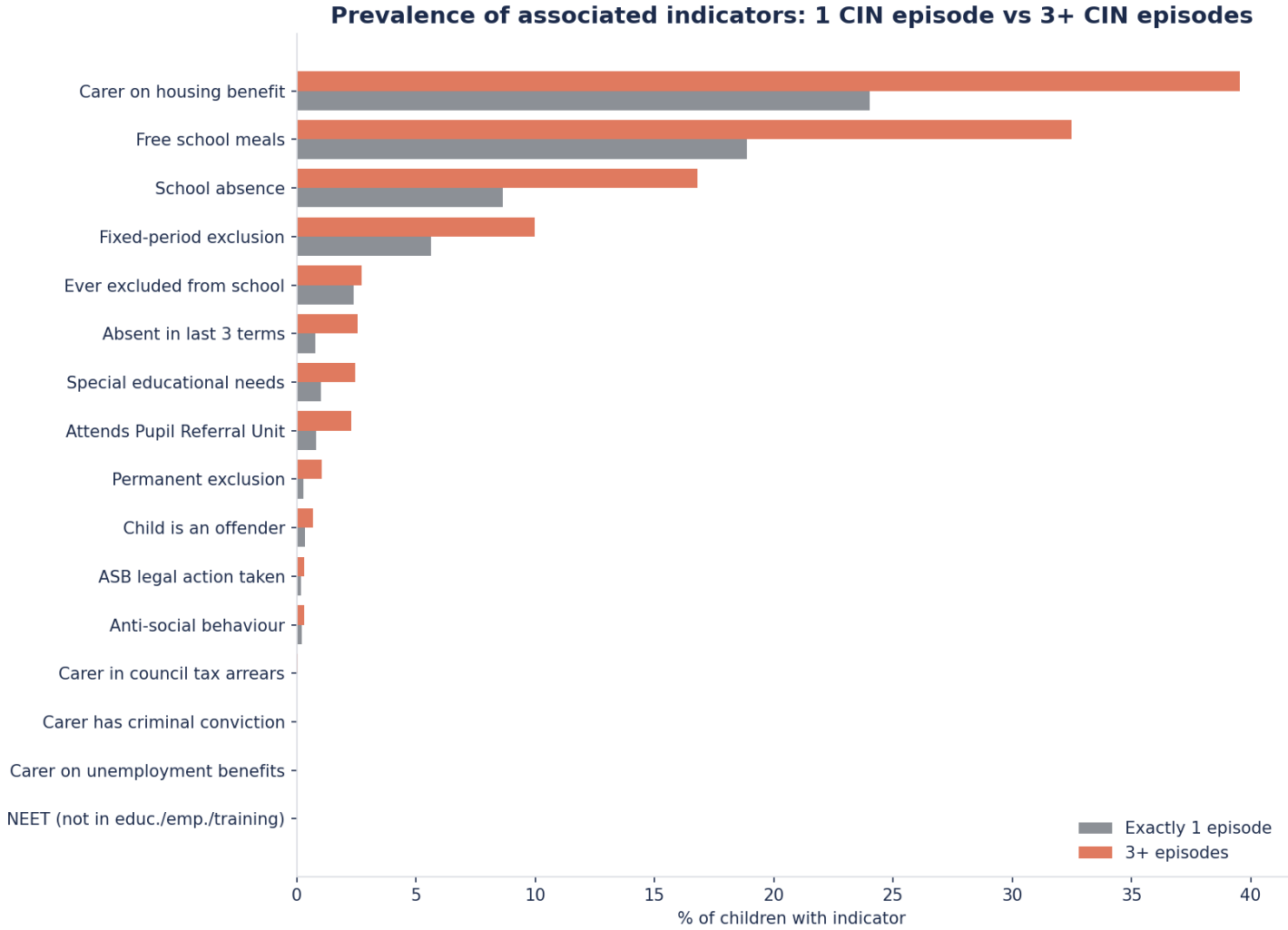
Housing benefit

+13.6pp

Free school meals

+8.2pp

School absence



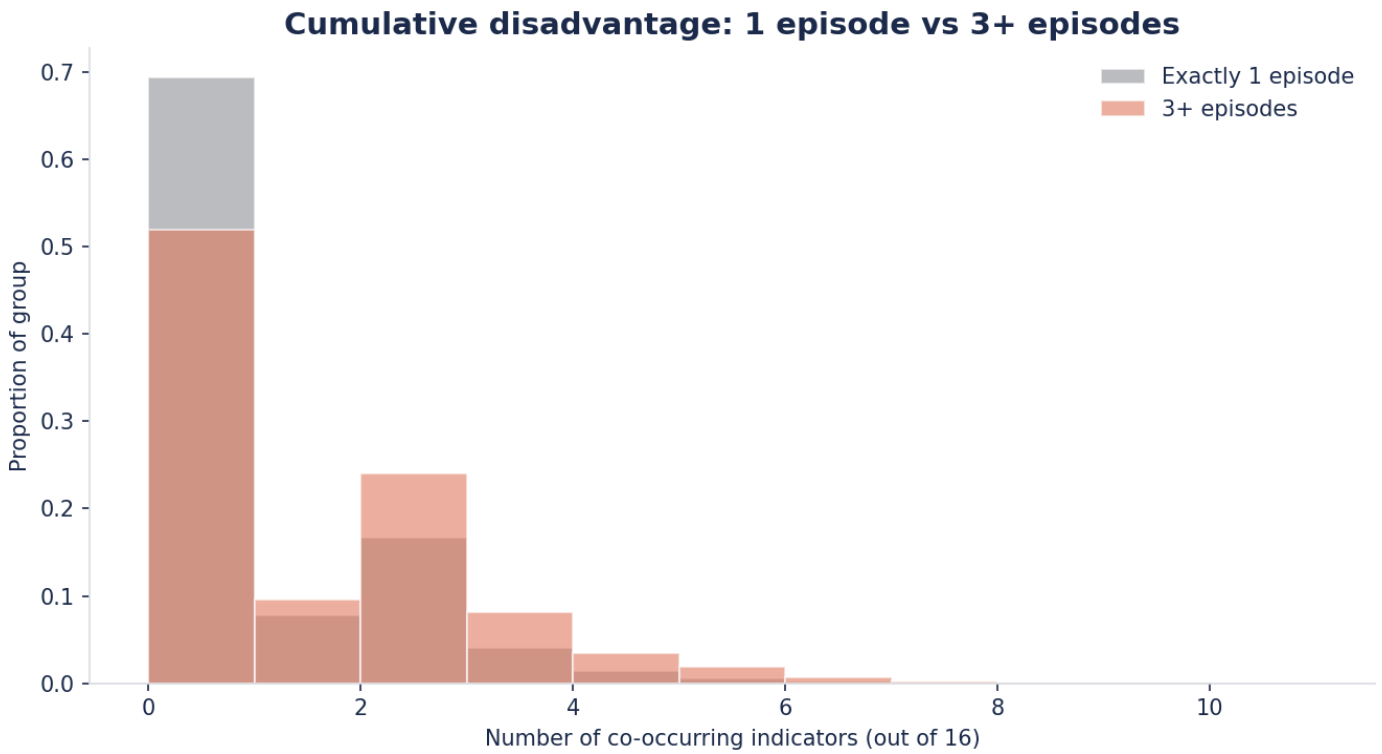
All differences shown are statistically significant (Fisher's exact test, p<0.01) except the bottom five indicators on the chart.

The gaps behind the chart

Indicator	1 episode	3+ episodes	Gap	p-value
Carer on housing benefit	24.0%	39.6%	+15.5pp	<0.001
Free school meals	18.9%	32.5%	+13.6pp	<0.001
School absence	8.7%	16.8%	+8.1pp	<0.001
Fixed-period exclusion	5.6%	10.0%	+4.3pp	<0.001
Absent in last 3 terms	0.8%	2.5%	+1.8pp	<0.001
Attends Pupil Referral Unit	0.8%	2.3%	+1.5pp	<0.001
Special educational needs	1.0%	2.5%	+1.4pp	<0.001
Permanent exclusion	0.3%	1.0%	+0.7pp	<0.001
Child is an offender	0.3%	0.7%	+0.3pp	0.009
Ever excluded from school	2.4%	2.7%	+0.3pp	0.262 (n.s.)

n.s. = not statistically significant at the 5% level. Test: Fisher's exact test (chosen over chi-square because several indicators are rare, and chi-square's approximation is unreliable with small counts).

Repeat episodes rarely arrive alone — indicators tend to stack

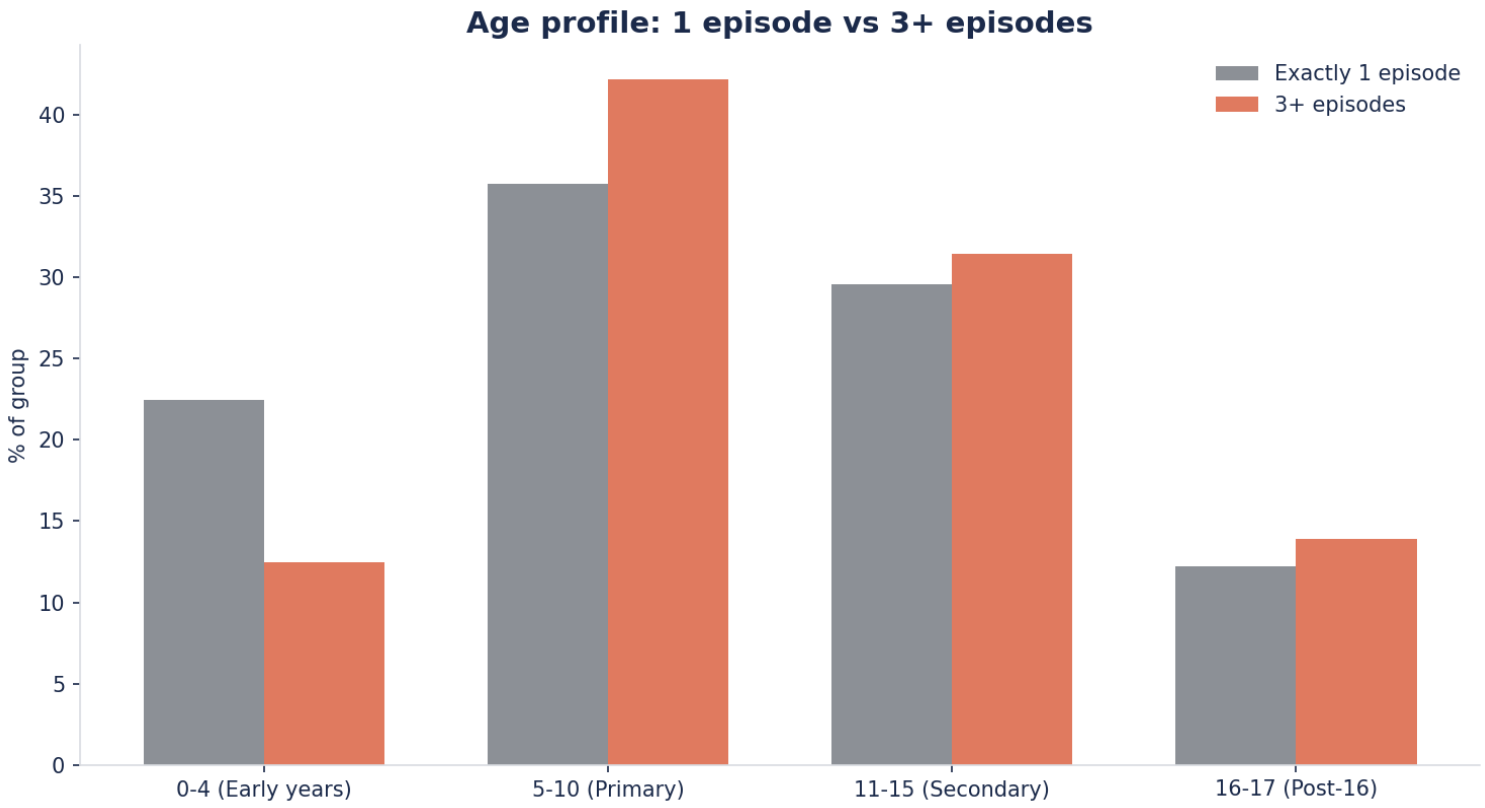


1.8×

more co-occurring indicators, on average, for children with 3+ CIN episodes (1.11) vs. exactly 1 (0.63). This is a descriptive count, not a validated risk score.

This supports thinking of repeat CIN involvement less as an isolated event and more as one marker within a wider cluster of household-level disadvantage.

Under-5s rarely recur; primary-age children make up the bulk of repeats



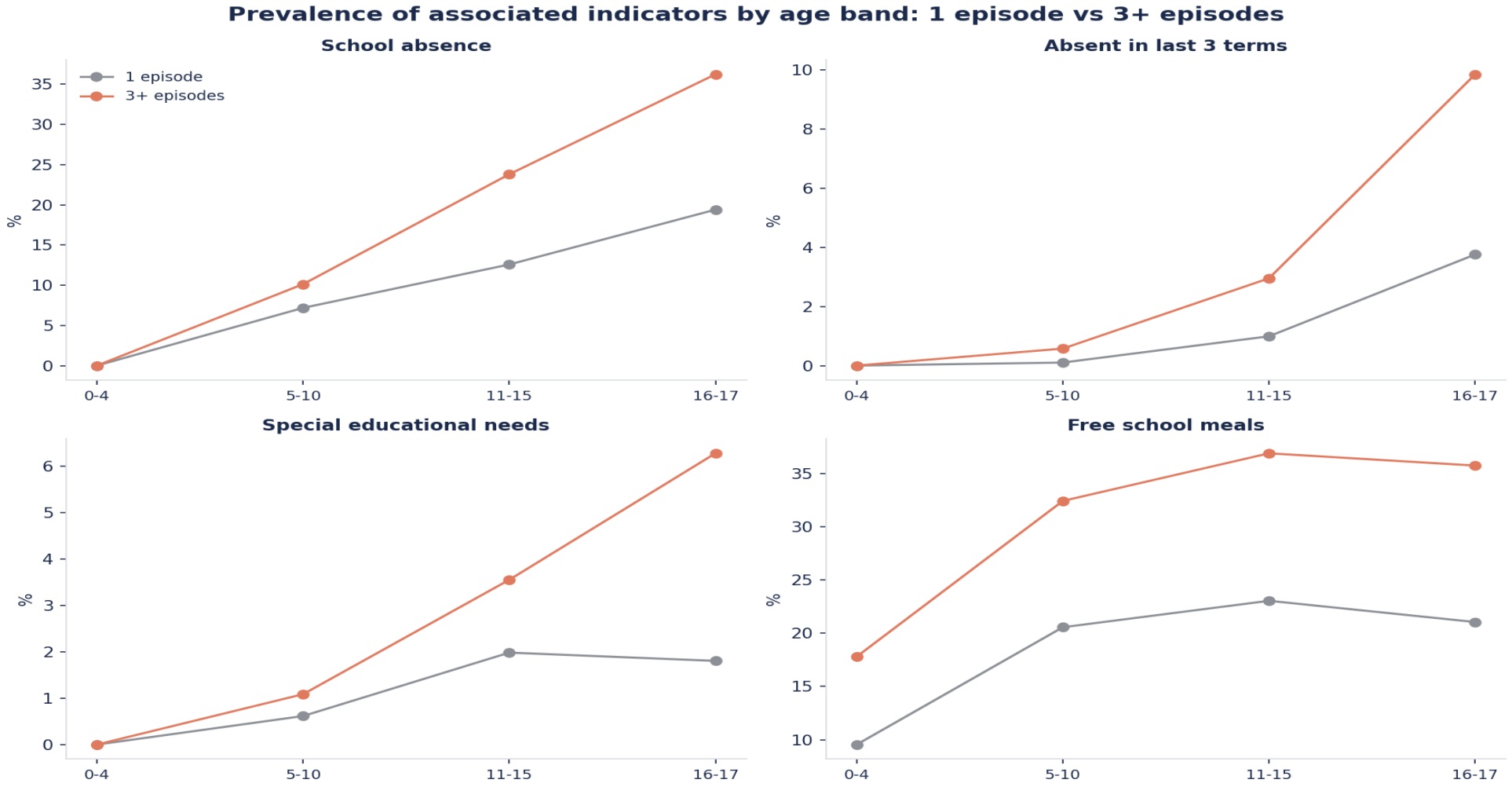
WHAT STANDS OUT

22.5% of the single-episode group is aged 0-4, vs. only **12.5%** of the 3+ episode group.

Very young children's CIN involvement looks more likely to resolve after one episode. Recurrence is concentrated in primary-school age (42% of the 3+ group is aged 5-10) and continues into secondary school.

Mean age is 9.2 (1 episode) vs. 9.9 (3+ episodes) — a small but statistically significant gap (p<0.001).

The size of the gap itself changes with age



SUMMARY

Four things we'd want the Local Authority to take away

01

Repeat need clusters with disadvantage

Housing benefit, free school meals and school absence are the strongest markers separating single-episode from repeat-episode children.

02

Indicators rarely come alone

Children with 3+ episodes carry 1.8× as many co-occurring indicators — repeat CIN involvement is a signal of wider household pressure, not an isolated issue.

03

Age changes the story

Under-5s look more likely to resolve after one episode; recurrence is concentrated in primary and secondary years.

04

Attendance gaps widen with age

The absence-related gap between groups grows sharply through the teenage years — a plausible window for earlier intervention.

What this analysis can’t tell us — yet

No timing information

We can't tell whether indicators (e.g. absence) preceded or followed a CIN episode — so we can describe association, not cause and effect.

Unpopulated fields

DWP, AdultOffender and NEET carry no signal in this extract and were excluded rather than reported as “0% prevalence.”

No demographic or geographic detail

No gender, ethnicity, ward or deprivation data, which limits any equity or targeting analysis.

Single Local Authority

Findings reflect one LA's population, thresholds and recording practices — they may not generalise elsewhere.

Other datasets we'd want, given no constraints

Case & referral history

- MASH / contact & referral records, incl. reason for referral and step-up/step-down dates
- Child Protection Plan history (starts, ends, categories)
- Social worker / caseload continuity

Household context

- Parental mental health, substance misuse & domestic abuse indicators (via multi-agency safeguarding data)
- Housing type, moves, and area deprivation (IMD) by ward
- Family structure & sibling CIN involvement

Outcomes & wider services

- Health contacts (A&E, GP, health visitor)
- School attainment & SEN provision detail, not just a flag
- Geography, ethnicity & gender for equity analysis

What we'd ask the Children Social Care Manager

1. When a case reopens, is that usually a new concern, or the same underlying issue resurfacing? Is that distinction currently recorded?
2. Is there a working definition of when an episode “ends” — and a target time before a case should be considered for step-down?
3. Do repeat episodes reflect families known to be higher-need from the outset, or does something change mid-way through casework?
4. What thresholds currently move a case from CIN into statutory Child Protection — and how many of our 3+ group have crossed that line?
5. Are there known capacity or caseload pressures around specific ages (e.g. transition to secondary school) that could shape the patterns we're seeing?

1

Watch the housing benefit / FSM / attendance signal

These three factors show the widest gaps between single- and repeat-episode children — a natural starting point for an early-warning indicator.

2

Focus early support on primary-age children

Recurrence is heaviest in the 5-10 age band; the 0-4 group's episodes look more likely to resolve without repeat involvement.

3

Clarify data recording before building on it further

Fix the exclusion-flag inconsistency and confirm why DWP/AdultOffender/NEET are unpopulated, so future analysis — including any predictive model — rests on reliable fields.

Thank you

Looking forward to discussing this with the Social Finance team.

Code and full analysis output accompany this deck.